

Kalo Kupuna

Setup

```
### Make sure to run the Google Sheet code first to authenticate.

if (!("pacman" %in% installed.packages()[, 1])) {

  install.packages("pacman")

} else {

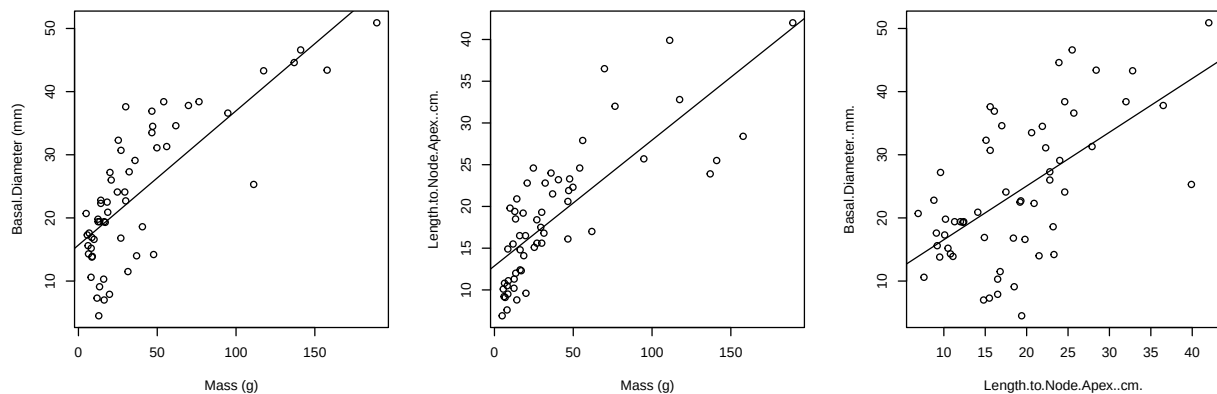
  library(pacman)
  p_load(rvest, magrittr, wordcloud, RColorBrewer, rmarkdown, bipartite,
        igraph, googlesheets4, knitr, xtable, reshape2)

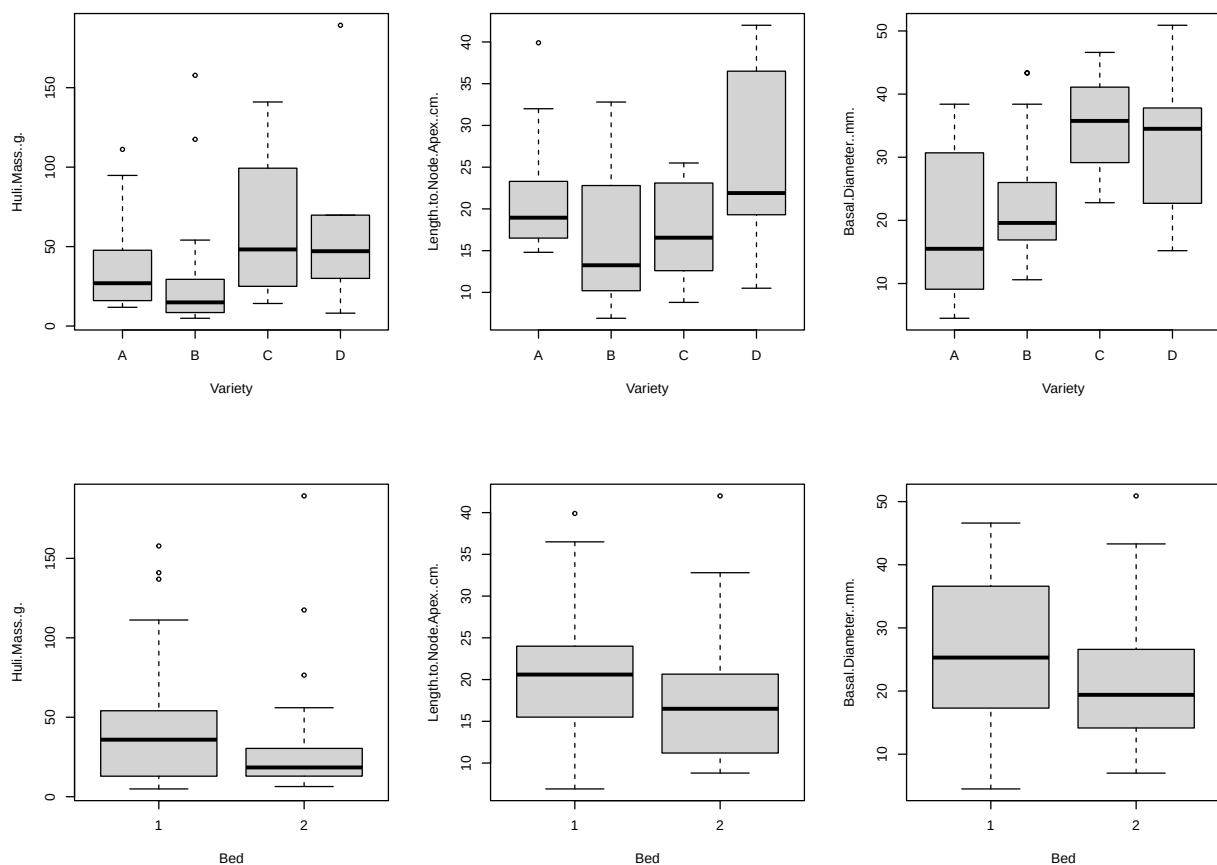
}
```

Mon Apr 14 14:52:02 HST 2025

Analyzing huli data

```
## v Reading from "Kalo_Measurements".
## v Range ''28Mar2025''.
## v Reading from "Kalo_Measurements".
## v Range ''Map''.
```





	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	15.6997455	1.291944	12.152035	0
huli[, "Huli.Mass..g"]	0.2126223	0.022794	9.327975	0

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.8760401	0.9064460	14.204973	0
huli[, "Huli.Mass..g"]	0.1506905	0.0159926	9.422497	0

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.0356951	3.1065145	2.586724	0.0123648
huli[, "Length.to.Node.Apex..cm"]	0.8509638	0.1526885	5.573202	0.0000008

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(huli[, "Huli.Variety"])	3	12688.64	4229.547	2.68033	0.0561446
Residuals	53	83633.73	1577.995	NA	NA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(huli[, "Huli.Variety"])	3	571.5389	190.51297	3.398947	0.0242861
Residuals	53	2970.6814	56.05059	NA	NA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(huli[, "Huli.Variety"])	3	1878.817	626.27248	6.348633	0.0009299
Residuals	53	5228.282	98.64683	NA	NA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(huli[, "Bed"])	1	3375.627	3375.627	1.997482	0.1631956
Residuals	55	92946.748	1689.941	NA	NA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(huli[, "Bed"])	1	64.75355	64.75355	1.024149	0.3159683
Residuals	55	3477.46680	63.22667	NA	NA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(huli[, "Bed"])	1	371.0296	371.0296	3.029457	0.0873578
Residuals	55	6736.0697	122.4740	NA	NA

Web Scrape Kupuna Kalo Website

```
extract_var_info <- function(x){
  x <- x[[8]]
  x <- strsplit(x, "\n\n")[[1]]
  main <- c(x[1], x[2])
  main[1] <- gsub("\n001\n\\*", "", main[1])
  x <- x[-1:-2]
  headers <- c("USE AS FOOD", "DISTRIBUTION", "GENERAL CHARACTERISTICS",
               "HA (PETIOLE)", "LAU OR LU'AU(LEAF BLADE)", "I'O KALO (CORM)",
               "PUA (FLOWER)", "REMARKS")
  ## fix missing sections
  fix.section <- sapply(headers, function(h, s, x)
    x[which(x == h) + 1] %in% s,
    s = sections, x = x)
  fix.section <- match(names(fix.section[fix.section]), x)
  for (i in seq_along(fix.section)){
    x <- c(x[1:fix.section[i]], "NA", x[(fix.section[i] + 1):length(x)])
  }
  x <- x[!(grepl("no items found", x, ignore.case = TRUE))]
  x <- x[!(grepl("bulletin 84 system", x, ignore.case = TRUE))]
  info <- x[(1:length(x)) %% 2 == 0]
  if (length(info) < length(headers)){
    info <- c(info, rep("NA", length(headers) - length(info)))
  }
}
```

```

}
out <- data.frame(sections = headers,
                  info = info)
return(out)
}

html <- read_html("https://www.kupunakalo.com/kalo-varieties")
txt <- html %>% html_elements("div") %>% html_text2()
var <- strsplit(txt[[14]], "\n") %>% unlist()
var <- var[!(grepl("choose", var, ignore.case = TRUE))]
var <- tolower(var)
var <- sapply(var, function(str) gsub("^'", "", str))
var <- sapply(var, function(str) gsub("[^[:alnum:]]", "-", str))
var <- iconv(var, from="UTF-8", to="ASCII//TRANSLIT")
var <- gsub(" ", "-", var)
var <- gsub("--", "-", var)
url <- paste0("https://www.kupunakalo.com/kalo-variety/",
              var)
html.var <- lapply(url, read_html)
txt.var <- lapply(html.var, function(html) html %>% html_elements("div") %>% html_text2())
info.var <- lapply(txt.var, extract_var_info)

ike.kk <- do.call(rbind, lapply(info.var, function(x) x[, 2]))
colnames(ike.kk) <- info.var[[1]][, 1]

info.fw <- paste(ike.kk[, "USE AS FOOD"], collapse = " ")
info.fw <- tolower(info.fw)
info.fw <- gsub("\\.", "", info.fw)
info.fw <- gsub(",", "", info.fw)
info.fw <- gsub("-", "", info.fw)
info.fw <- gsub(";", "", info.fw)
info.fw <- gsub(":", "", info.fw)
info.fw <- gsub("\\(", "", info.fw)
info.fw <- gsub("\\)", "", info.fw)
info.fw <- strsplit(info.fw, split = " ")[[1]]
info.fw <- info.fw[is.na(match(info.fw, c("by", "from", "to", "or", "also", "the",
    "for", "in", "on", "as", "a", "an", "it",
    "are", "is", "and", "of")))]
tab.fw <- table(info.fw)

pal <- brewer.pal(6, "Dark2")
pal <- pal[-(1)]
wordcloud(names(tab.fw), tab.fw, colors = pal,
          vfont=c("gothic english", "plain"))

srch.red <- (grepl("red", tolower(ike.kk[, "USE AS FOOD"])) |
            grepl("lehua", tolower(ike.kk[, "USE AS FOOD"])))
var[srch.red]

srch.ala <- (grepl("ala", tolower(ike.kk[, "USE AS FOOD"])) |
            grepl("'ala", tolower(ike.kk[, "USE AS FOOD"])) |
            grepl("'āla", tolower(ike.kk[, "USE AS FOOD"])) |
            grepl("āla", tolower(ike.kk[, "USE AS FOOD"])))

```

```
var[srch.ala]

srch.ulika <- (grepl("ulika", tolower(ike.kk[, "USE AS FOOD"]))) |
  grepl("gummy", tolower(ike.kk[, "USE AS FOOD"])))
var[srch.ulika]
```

Kalo Kupuna – Group Key

- 1a. Leaf margine undulating [2]
- 1b. Leaf margine without distinct undulations [3]
- 2a. Mature leaves ovate, often concave [Kai]
- 2b. Mature leaves saggitate and long relative to width [Lauloa]
- 3a. Leaf sinus connecting to piko, distinct inner petiole ridge [Piko]
- 3b. Leaf sinus shallow and/or no distince petiolar ridge [4]
- 4a. Petioles green, no distinct patterning, leaves saggitate [Lehua]
- 4b. Petioles with mixed colors or patterned or with dark shades, especially red [5]
- 5a. Petioles striped either red-purple on green or green on red-purple, leaves narrow/sagittate [Manini]
- 5b. Petioles not green-red/purple striped or not sagittate leaves [6]
- 6a. Petioles red to purple becoming lighter red/pink near kohina, leaves narrow and small relative to petiole [Ulaula]
- 6b. Colors light or leaves larger and/or wider and/or petioles with darker colors and distinct pattern near kohina [7]
- 7a. Color lighter (not black, red-purple, red-brown), leaves small and verticle, corms branching [Mana]
- 7b. Petioles black to red-purple and/or red-brown with distinct distinct lilac-red rings 3-4 cm above kohina [Eleele]

Japanese

Palauan

Tongan

Samoan

Tahitian

Chinese

Hawaiian

GROUP MANA GROUP PIKO GROUP KAI GROUP LAULOA GROUP ELEELE GROUP ULAULA
GROUP MANINI GROUP LEHUA

Group Reported Extant Māna 16 5 Lauloa 13 5 Lehua 11 4 Kāi/'Ala 12 4 Manini 9 5 Piko/Haehae/Uaua 7 6
Hā-o-kea 6 1 Hāpu'u(pu'u) 6 1 Kūmū/'Ula'ula 6 3 Naioea/Nohu/'Ele'ele 6 2 Total 92 46

Asian Kāi Palauan Mana 'Ula'ula Lauloa-Manini

Accession No. Traditional classification Revised classification Geographic Origin 'Iliuaua ("Pake") 4 — Asian
? Bun-long 5 — Asian China 'Āweu ("wild taro") 6 — — Hawaii'i Kakakura 7 — 'Ula'ula South Pacific
Mana 'Ulu 8 Mana Mana Hawaii'i Mana 'Ōpelu 9 Mana Mana Hawaii'i Mana Uliuli 11 Mana Mana South
Pacific Mana 'Ula'ula 12 Mana Mana Hawaii'i Mana Lauloa 13 Mana Mana Hawaii'i Mana Ke'oke'o 14 Mana
Mana Hawaii'i Piko Lehua Apei 16 Piko Mana Hawaii'i Piko Kea 18 Piko Lehua Hawaii'i Piko Ke'oke'o 19

Piko Lehua Hawai'i Piko Uaua 20 Piko Lehua Hawai'i Piko Uliuli 21 Piko Lehua Hawai'i Piko 'Ele'ele 22
 Piko Lauloa-Manini Hawai'i 'Elepaio Hā Kea 23 — Lehua Hawai'i Uahiapele 24 — Lauloa-Manini Hawai'i
 Manapiko 25 — Mana Hawai'i Tahitian 26 — Mana South Pacific Kāi Uliuli 27 Kāi Kāi Hawai'i Kāi Kea 29
 Kāi Kāi Hawai'i 'Apuwai 30 — Lehua Hawai'i Pi'i'ali'i 32 — Lehua Hawai'i Pa'akai 33 — Lehua Hawai'i
 Moana 34 — Lehua Hawai'i Akuugawai 35 — Lauloa-Manini South Pacific Lauloa 'Ele'ele 'Ōma'o 36 Lauloa
 Lauloa-Manini Hawai'i Lauloa 'Ele'ele 'Ula 37 Lauloa Lauloa-Manini Hawai'i Lauloa Palakea 'Ele'ele 38
 Lauloa Lauloa-Manini Hawai'i Lauloa Palakea 'Ula 39 Lauloa Lauloa-Manini Hawai'i Lauloa Palakea Papamū
 40 Lauloa Lauloa-Manini Hawai'i Lauloa Palakea Ke'oke'o 41 Lauloa Lehua Hawai'i Lauloa Ke'oke'o 42
 Lauloa Lehua Hawai'i 'Ele'ele Mākoko 43 'Ele'ele Lehua Hawai'i 'Ele'ele Naioea 44 'Ele'ele Lehua Hawai'i
 Manini 'Ōwali 45 — Lauloa-Manini Hawai'i Kūmū 'Ele'ele 46 — Lauloa-Manini Hawai'i Nāwao 47 — Lehua
 Hawai'i 'Ula'ula Kūmū 48 'Ula'ula 'Ula'ula Hawai'i 'Ula'ula Poni 49 'Ula'ula 'Ula'ula Hawai'i 'Ula'ula Moano
 50 'Ula'ula 'Ula'ula Hawai'i Niue 'Ula'ula 51 — 'Ula'ula South Pacific 'O'opukai 52 — 'Ula'ula Hawai'i Manini
 Uliuli 53 Manini Lauloa-Manini Hawai'i Manini Kea 54 Manini Lauloa-Manini Hawai'i Pāpākolea Koa'e 56 —
 Lehua Hawai'i Nihopu'u 58 — Lauloa-Manini Hawai'i Manini 'Ōpelu 59 — Lauloa-Manini Hawai'i Lehua
 Ke'oke'o 64 Lehua Lehua Hawai'i Lehua 'Ele'ele 65 Lehua Lehua Hawai'i Lehua Pala'i'i 66 Lehua Lehua
 Hawai'i 'Apowale 67 — Lehua Hawai'i Wehiwa 68 — Lehua Hawai'i Papapueo 69 — Lehua Hawai'i Kū'oho
 70 — Lehua Hawai'i Māea 72 — Lehua Hawai'i Haokea 73 — Lehua Hawai'i Hāpu'u 75 — Lehua Hawai'i
 False Lihilihimōlina 78 — — Hawai'i Mana 'Okō'a 80 — Mana Hawai'i Moi 81 — Lehua Hawai'i Pololū 84 —
 Lehua Hawai'i Moi 'Ula'ula n/a n/a Lehua Hawai'i Makalau (Moloka'i) n/a n/a Mana Hawai'i Ngesuas (P-1)
 n/a n/a Palauan Palau Ochelochel (P-7) n/a n/a Palauan Palau Ngeruuch (P-10) n/a n/a Palauan Palau
 Merii (P-12) n/a n/a Palauan Palau Dirratengadik (P-20) n/a n/a Palauan Palau